

PART A

(25 marks)

1. Evaluate the following limits.

(a) $\lim_{x \rightarrow 16} \frac{\sqrt{x} - 4}{x - 16}$

(2 marks)

(b) $\lim_{x \rightarrow -\infty} \frac{3x + |1 - 3x|}{1 - 5x}$

(3 marks)

(c) Find constants a and b such that $\lim_{x \rightarrow 0} \frac{\sqrt{ax+b} - 2}{x} = 1$

(6 marks)

2. (a) Find the derivative of $f(x)$ using the first principle.
 $f(x) = x^2 - 3x$.

(4 marks)

- (b) Suppose the function g is defined by

$$g(x) = \begin{cases} k\sqrt{x+1} & \text{for } 0 \leq x \leq 3 \\ mx + 2 & \text{for } 3 < x \leq 5, \end{cases}$$

where k and m are constants. If g is differentiable at $x = 3$, what are the value of k and m ?

(5 marks)

3. Given the function $f(x) = x^3 e^{-x+5}$. The first and second derivatives are:

$$f'(x) = x^2(3-x)e^{-x+5} \quad \text{and} \quad f''(x) = x(x^2 - 6x + 6)e^{-x+5}$$

Determine the critical point(s) of f . Classify the critical point(s) as local maxima or local minima or neither.

(5 marks)

PART B*(75 marks)*

1. Given $z=a+bi$ is a complex number such that $|z|=5$,

(a) Show that $\frac{1}{5+z} = \frac{1}{10} - \frac{b}{10(5+a)}i$.

(4 marks)

(b) Hence, express $\frac{1}{5+z}$ in polar form with $a=3$.

(5 marks)

2. (a) Find the set of $\left| \frac{3x+5}{2} \right| < |1+x|$.

(6 marks)

(b) (i) Given that $\log_2 x + 2\log_4 y = 4$, show that $xy = 16$

(3 marks)

(ii) Hence, solve the simultaneous equation below such that $x > y$.

$$\log_{10}(x+y) = 1$$

$$\log_2 x + 2\log_4 y = 4$$

(3 marks)

3. (a) Given a function $f(x) = 2e^x + 1$.

(i) Show that f is a one-to-one function.

(2 marks)

(ii) State the domain and range of f .

(2 marks)

(iii) Determine the inverse function of f and state its domain and range.

(4 marks)

(iv) Sketch the graph of f and f^{-1} on the same axis.

(3 marks)

- (b) Find the numerical values of x when $g \circ f = g^2 \circ f$ given that

$$f : x \rightarrow 3x - 5 \quad \text{and} \quad g : x \rightarrow \frac{1}{2x} + 1.$$

(7 marks)

4. (a) Evaluate $\lim_{x \rightarrow 5} \frac{\left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{5}} \right)}{x - 5}.$

(5 marks)

(b) The function f is define by $f(x) = \begin{cases} mx^2 - 3, & x < 2 \\ mx + n, & 2 \leq x < 7 \\ 3x - 3m, & x \geq 7 \end{cases}$

- (i) Find the values of m and n if the limit exists at $x = 2$ and $x = 7$.

(5 marks)

- (ii) At what values of $f(7)$ is the limit continuous?

(1 marks)

5. Find the derivatives of the following functions:

(a) $y = x^{\ln x}$

(4 marks)

(b) If $y = \frac{\sqrt{x^2+1} - \sqrt{x^2-1}}{\sqrt{x^2+1} + \sqrt{x^2-1}}$, show that $\frac{dy}{dx} = 2x - \frac{2x^3}{\sqrt{x^4-1}}.$

(4 marks)

6. The parametric equations for a curve **C** are given by

$$x = e^{-t}, \quad y = 2^{2t}, \quad \text{where} \quad -\infty < t < \infty.$$

- (a) Calculate $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ as a function of t directly from these equations.

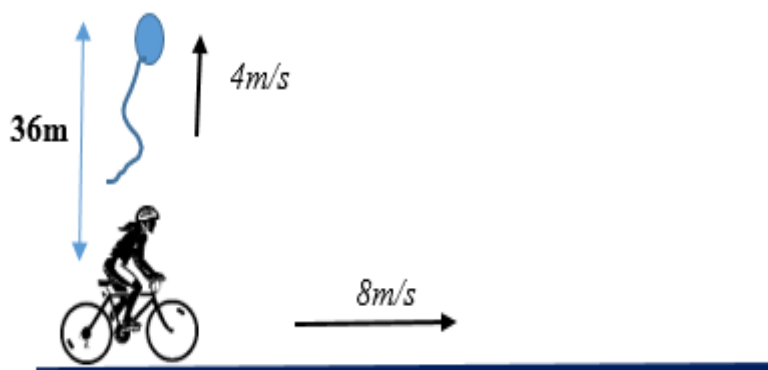
(6 marks)

- (b) Obtain the equation of the curve **C** in the form of $y = f(x)$

(3 marks)

7. A balloon is rising at a constant speed 4 m/s . A boy is cycling along a straight road at a speed of 8 m/s . When he passes under the balloon, it is 36 meters above him. How fast is the distance between the boy and balloon increasing 3 second later? (Using *Pythagoras Theorem* in your solution)

(8 marks)



END OF QUESTIONS PAPER

Answers (Part A)

1(a) $\frac{1}{8}$

(b) 0

(c) $a=4, b=4$

2(a) $2x-3$

(b) $k = \frac{8}{5}; m = \frac{2}{5}$

- 3 $x=0, x=3$ are critical points
 $(0,0)$ neither a local maximum
 nor local minimum.
 $(3, 27e^2)$ is a local maximum.

Answers (Part B)

1(a) Shown

(b) $\frac{\sqrt{5}}{20} [\cos(-0.4636\text{rad}) + i \sin(-0.4636\text{rad})]$

2(a) $\{x: -3 < x < -1.4\}$

(b) (i) shown

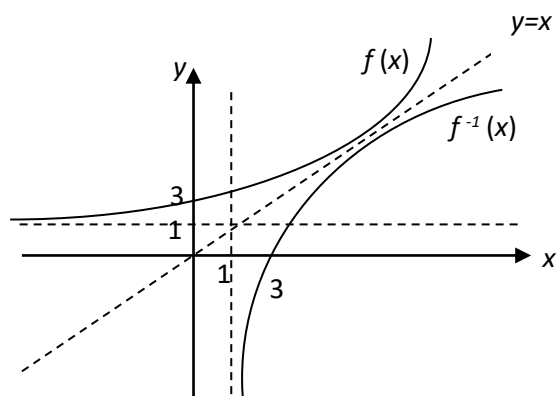
(ii) $x=8, y=2$

3(a) (i) shown

(ii) $D_{f(x)} = (-\infty, \infty), R_{f(x)} = (1, \infty)$

(iii) $D_{f^{-1}(x)} = (1, \infty), R_{f^{-1}(x)} = (-\infty, \infty)$

(iv)



(b) $x = 2.122$ or $x = 1.545$

4(a) $\frac{1}{10\sqrt{5}}$ or $-\frac{\sqrt{5}}{50}$

(b)

(i) $m=2, n=1$

(ii) 15

$$5(a) \quad \frac{dy}{dx} = 2(\ln x) x^{\ln x - 1}$$

(b) Shown

$$6(a) \quad \frac{dy}{dx} = e^t 2^{2t+1} \ln 2 ; \quad \frac{d^2 y}{dx^2} = e^{2t} 2^{2t+1} (1 + 2 \ln 2) \ln 2$$

$$(b) \quad y = 2^{-2 \ln x}$$

$$7 \quad \frac{ds}{dt} = \frac{16}{\sqrt{5}} m/s$$